POINTS OF FINITE TYPE ON AN ALGEBRAIC SPACE CHARACTERIZED BY ETALE PRESENTATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof uses the notion of a finite type point of a scheme, which is defined in the Morphisms chapter; the shrinking step at the end is justified by the cited identification of finite type points with closed points after shrinking.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The general principle translating representable properties into scheme properties. Spaces, Lemma spaces-lemma-representable-transformations-property-implication, used at proof lines 5035-5037. Makes U -> X locally of finite type.
- Etale morphisms are locally of finite presentation, hence locally of finite type. Morphisms, Lemmas morphisms-lemma-etale-locally-finite-presentation and morphisms-lemma-finite-presentation-finite-type, used at proof lines 5038-5040. Gives local finite typeness of the etale morphisms involved.
- Permanence of local finite typeness under composition. Lemma lemma-composition-finite-type, used at proof lines 5040-5042. Produces the locally finite type representative in the forward direction.
- Finite type points of schemes and the identification of finite type with closed after shrinking. Morphisms, Lemmas morphisms-lemma-identify-finite-type-points and morphisms-lemma-finite-type-points-morphism, used at proof lines 5048-5056. Provides the finite type point v and turns its image into a closed point.

Source reference. Original lemma 03XG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5147–5147. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3238–3238. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2389–2389. Navigation pointer into the Stacks

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02J3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3322–3322. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02J2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3321–3321. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3883–3883. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$. The following are equivalent:*

1. *There exists a morphism $\mathrm{Spec}(k) \to X$ which is locally of finite type and represents x.*
2. *There exists a scheme U, a closed point $u \in U$, and an étale morphism $\varphi : U \to X$ such that $\varphi(u) = x$.*

Proof. Let $u \in U$ and $U \to X$ be as in (2). Then $\mathrm{Spec}(\kappa(u)) \to U$ is of finite type, and $U \to X$ is representable and locally of finite type (by the general principle Spaces, Lemma 02YO and Morphisms, Lemmas 02GR and 01TW). Hence we see (1) holds by Lemma 03XG.

Conversely, assume $\mathrm{Spec}(k) \to X$ is locally of finite type and represents x. Let $U \to X$ be a surjective étale morphism where U is a scheme. By assumption $U \times_X \mathrm{Spec}(k) \to U$ is locally of finite type. Pick a finite type point v of $U \times_X \mathrm{Spec}(k)$ (there exists at least one, see Morphisms, Lemma 02J2). By Morphisms, Lemma 02J3 the image $u \in U$ of v is a finite type point of U. Hence by Morphisms, Lemma 02J2 after shrinking U we may assume that u is a closed point of U, i.e., (2) holds. □